



Lunar regolith simulant-based triboelectric nanogenerators via solid-state additive manufacturing

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ABSTRACT

For future lunar missions, in-situ resource utilization (ISRU) is essential to reduce reliance on Earth-based supply chains and enable sustained human presence beyond Earth. Realizing this vision critically depends on the efficient on-site energy harvesting from locally available extraterrestrial materials, such as lunar regolith. Utilizing regolith as a functional material can enable autonomous, self-powered systems for sustained surface operations while reducing logistical complexity. In parallel with this vision, this study develops lunar regolith simulant-based triboelectric nanogenerators (TENGs) using a solid-state additive manufacturing approach. Engineered regolith/metal composite powders (50–80 wt% regolith) are synthesized and deposited onto polytetrafluoroethylene (PTFE) substrate using cold spray additive manufacturing (CSAM) to fabricate high-performance, mechanically robust triboelectric layers. The resulting layers exhibit electrical conductivity up to $\sim 4.9 \times 10^5 \text{ S.m}^{-1}$, with thicknesses ranging from 16 to 40 μm . Furthermore, radiation testing using a high-power linear accelerator confirmed the radiation resistance of the fabricated triboelectric layers, along with their strong interfacial adhesion strength. Notably, under cyclic contact-separation loading, the lunar-based TENG (with 50 wt% regolith) achieves an open-circuit voltage (V_{oc}) of 142 V and a short-circuit current (I_{sc}) of 17.6 μA . Moreover, a power density of 81.3 $\mu\text{W/cm}^2$ is obtained, representing a significant improvement (~ 27 -fold) over previously reported lunar regolith-based TENGs. The fabricated TENGs also maintain stable energy-harvesting performance over 5000 cycles, demonstrating excellent durability and reliability. This work presents the first proof-of-concept demonstration of CSAM-enabled lunar regolith-based TENGs with promising performance, offering a viable pathway toward durable, in-situ energy-harvesting systems for future lunar missions.

1. Introduction

The development of sustainable lunar infrastructure requires advances in in-situ resource utilization (ISRU) [1] to enable the fabrication of structural and functional components directly from locally available materials in extraterrestrial environments. Beyond structural elements, the ability to produce conductive surfaces (electrodes), electronic devices, and systems is critical for enabling energy-harvesting technologies that support autonomous, long-duration operations on the lunar habitat [2]. Particularly, as the vision of establishing a sustained human presence on the Moon advances, scalable and resource-efficient on-site manufacturing of energy and sensing systems becomes essential to reduce reliance on Earth-based supply chains [3,4]. In particular, extended lunar missions will require advanced spacesuits and systems

integrated with sensors and electronics for physiological monitoring, environmental sensing, navigation, communication, health diagnostics, and adaptive life-support systems [5,6]. The reliable operation of these technologies for future lunar missions depends on a stable and continuous power supply derived from locally available resources.

In this context, triboelectric nanogenerators (TENGs) offer a promising solution for lightweight, distributed energy harvesting by effectively converting ambient mechanical energy (e.g., motion, vibration, and contact) into electrical output, enabling self-powered sensing and reducing dependence on conventional power sources [7–10]. The working principle of TENG is based on contact electrification and electrostatic induction, where electron transfer occurs when two materials with different polarities come into contact and then separate or

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slide against each other, allowing energy to be harvested through an external circuit [11]. Owing to their lightweight structure, material versatility, low cost, and capability for self-powered operation, TENGs have emerged as promising technologies for wearable electronics, autonomous sensing systems, and energy-harvesting applications [12–15].

In lunar habitats, TENGs would be particularly attractive due to their simple architecture, low material requirements, and compatibility with flexible, conformal surfaces [16]. They can harvest energy from routine astronaut activities, structural vibrations, and regolith interactions, providing a continuous, decentralized power source for embedded sensors and low-power electronics [17]. Moreover, their lightweight and flexible design may enable seamless integration into spacesuits, habitat interiors, and surface infrastructure, supporting distributed sensing networks without adding significant mass or complexity. This decentralized energy generation may enhance system resilience and redundancy while reducing reliance on centralized power systems and frequent maintenance or resupply. Collectively, these advantages highlight the strong potential and growing importance of developing lunar regolith-based TENGs for future lunar missions.

In parallel with this vision, several recent studies have explored lunar regolith simulant-based TENGs, aiming at ISRU for energy harvesting applications [18,19]. Yohannan et al. [18] developed a flexible triboelectric electrode by embedding lunar regolith simulant into a polydimethylsiloxane (PDMS) matrix via casting (doctor blade) method. They later demonstrated an all-3D-printed TENG by incorporating regolith into a PLA matrix using fused filament fabrication (FFF), followed by PDMS infusion to create porous structure [19]. Despite these advances, both casting and FFF rely on phase-change processes (e.g., melting or solidification), which introduce significant constraints for lunar deployment, including the need for controlled thermal management, pressurized environments, and higher energy input. These requirements may limit their practicality for in-situ, on-site manufacturing under lunar conditions. As such, there remains a critical need for alternative fabrication strategies that are compatible with the vacuum, microgravity, and extreme temperature fluctuations of the lunar environment. Notably, given the performance of TENGs largely depends on the effectiveness of electrostatic interactions between the triboelectric layers and electrodes [11], there is also a strong need for highly durable and electrically conductive triboelectric architectures that can maintain stable performance under harsh operating conditions.

To contribute to this emerging field, this study employs cold spray additive manufacturing (CSAM) as a solvent-and binder-free approach for directly fabricating lunar regolith-based TENGs, as illustrated in Fig. 1a. CSAM is a solid-state process in which micron-sized powder particles are accelerated to supersonic velocities using a pressurized gas stream and deposited onto a substrate through high-velocity impact [20]. In this context, lunar highland simulant powder (LHS-1D) [21] is used as the primary feedstock and mechanically alloyed with metallic additives (Sn + Al_2O_3) through ball milling at various regolith weight ratios to synthesize composite feedstock powders for the CSAM process (see Fig. 1b). These powders are subsequently deposited onto polytetrafluoroethylene (PTFE) substrates to fabricate electrically conductive TENG electrodes. The fabricated electrodes are comprehensively characterized in terms of microstructure, surface morphology, electrical conductivity, mechanical properties, irradiation resistance, and adhesion strength. Furthermore, TENG devices with various electrode configurations are designed and evaluated through detailed energy-harvesting performance analyses, including demonstrations of energy harvesting and storage applications. Collectively, this study introduces the following key innovations:

- (i) The first proof-of-concept demonstration of the CSAM process for fabricating lunar regolith-based TENGs.
- (ii) Direct-write fabrication of electrically conductive, flexible, and durable triboelectric layers for TENG applications.

- (iii) Comprehensive characterization of the fabricated triboelectric layers (coatings) to establish CSAM process–structure–property relationships for enhanced TENG performance.
- (iv) Design and development of lunar regolith-based TENGs with various tribo-pair configurations to investigate their energy-harvesting performance.
- (v) Demonstration of energy harvesting, storage, and small-electronics applications using the fabricated TENGs.

The remainder of this work is organized as follows. Section 2 presents the materials and experimental methods. Section 3 provides the results and discussion, including feedstock powder characterization (Section 3.1), characterization of the fabricated coatings (Section 3.2), and the development of lunar regolith-based TENGs (Section 3.3). Finally, Section 4 summarizes the key findings and outlines future research directions.

2. Materials and methods

This section outlines the materials used, feedstock powder preparation, and subsequent solid-state deposition onto the target substrate via CSAM to form triboelectric layers. Fig. 1b presents the overall workflow from feedstock preparation to electrode fabrication for lunar regolith simulant-based functional layers for TENGs.

2.1. Lunar regolith simulant

Lunar highland simulant (LHS-1D, *Space Resource Technologies, USA* [21]) was used as the primary feedstock material. The LHS-1D simulant is engineered to closely replicate the chemical composition, mineralogy, and particle morphology of lunar highland soil [22]. It primarily consists of oxides: SiO_2 (49.12%), TiO_2 (0.63%), Al_2O_3 (26.29%), FeO (3.20%), MnO (0.06%), MgO (2.86%), CaO (13.52%), Na_2O (2.55%), K_2O (0.34%), P_2O_5 (0.17%), and LOI (0.41%) [21]. The composition and material properties of this material have been previously studied in Ref. [23]. Figs. 1b and 2a show the powder morphology of the LHS-1D simulant, with particle sizes ranging from ~ 0.04 to $35\ \mu\text{m}$. For the remainder of this work, the terms “lunar regolith simulant” and “lunar regolith” are used interchangeably.

2.2. Preparation of composite powder feedstocks

Lunar regolith is inherently electrically insulating due to its oxide-dominated composition and lack of free charge carriers, which limits its direct use in applications requiring electrical conductivity. This may limit the direct use as TENG electrodes due to reduced charge transfer efficiency and overall electrical performance. To address this limitation, the regolith was compositionally modified in this study by incorporating conductive metallic additives via the ball-milling process. Specifically, blended metallic additive powders comprising ductile Sn (55–45 wt%) and metal oxide Al_2O_3 (45–55 wt%) (i.e., both procured from *Titomic, Inc.*) were incorporated into the regolith matrix through ball milling at varying weight ratios (20–50 wt%). In the resulting composite powder, Sn serves as the primary conductive phase, while Al_2O_3 enhances CSAM deposition efficiency. Furthermore, given that lunar regolith inherently contains a significant fraction of Al_2O_3 (26.29 wt%), the engineered composite powder remains compatible with ISRU pathways, promising the potential for effective use of in-situ materials.

Noteworthy, although Sn is not a natural constituent of lunar regolith, its use in ISRU could be enabled through the recovery and recycling of mission-derived materials. Specifically, Sn can be extracted from solder alloys [24], printed circuit boards (PCBs) [25–27], electronic waste [28], and other end-of-life spacecraft components. While such recovery would require additional collection, separation, and processing infrastructure, these operations could become practical as

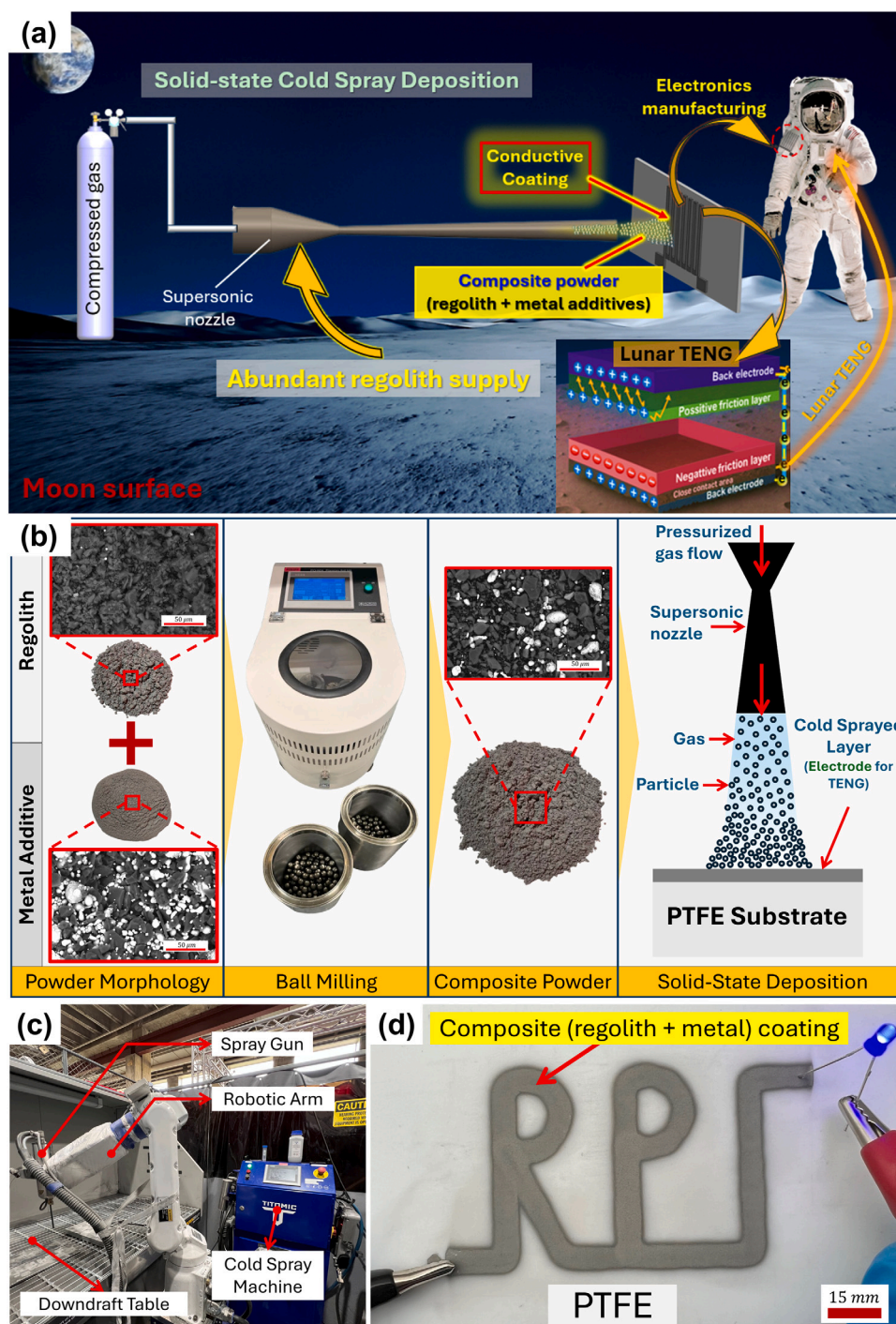


Fig. 1. CSAM of lunar regolith-based composite coatings (i.e., triboelectric layers): (a) Schematic illustration of CSAM-enabled fabrication of regolith-based TENGs; (b) Fabrication workflow of triboelectric layers on a PTFE substrate; (c) Robotic CSAM experimental setup; (d) Programmable electrically conductive coating on PTFE film.

sustained lunar activities generate larger material inventories and waste streams. Accordingly, the present Sn-regolith system represents a hybrid ISRU pathway, combining locally available regolith with a supplied or potentially recycled metallic phase. Future work should focus on reducing the required Sn fraction and exploring lunar-abundant or lunar-extractable ductile alternatives to progressively increase the utilization of locally sourced materials.

Ball milling was performed using a planetary mill (*Across International*) at a rotational speed of 220 rpm for 4 h without the use of

any solvent. A powder-to-ball weight ratio of 1:3 was maintained using stainless steel balls (6 mm diameter) in 100 mL jars. These parameters promoted homogeneous dispersion of the metallic constituents within the regolith matrix while minimizing particle fragmentation and contamination. Additionally, the ball-milling process facilitates mechanical alloying, which enhances ductility and, consequently improves deposition performance in the CSAM process. Overall, four different composite powders were synthesized with lunar regolith weight fractions ranging from 50 to 80 wt% to investigate the effect of regolith

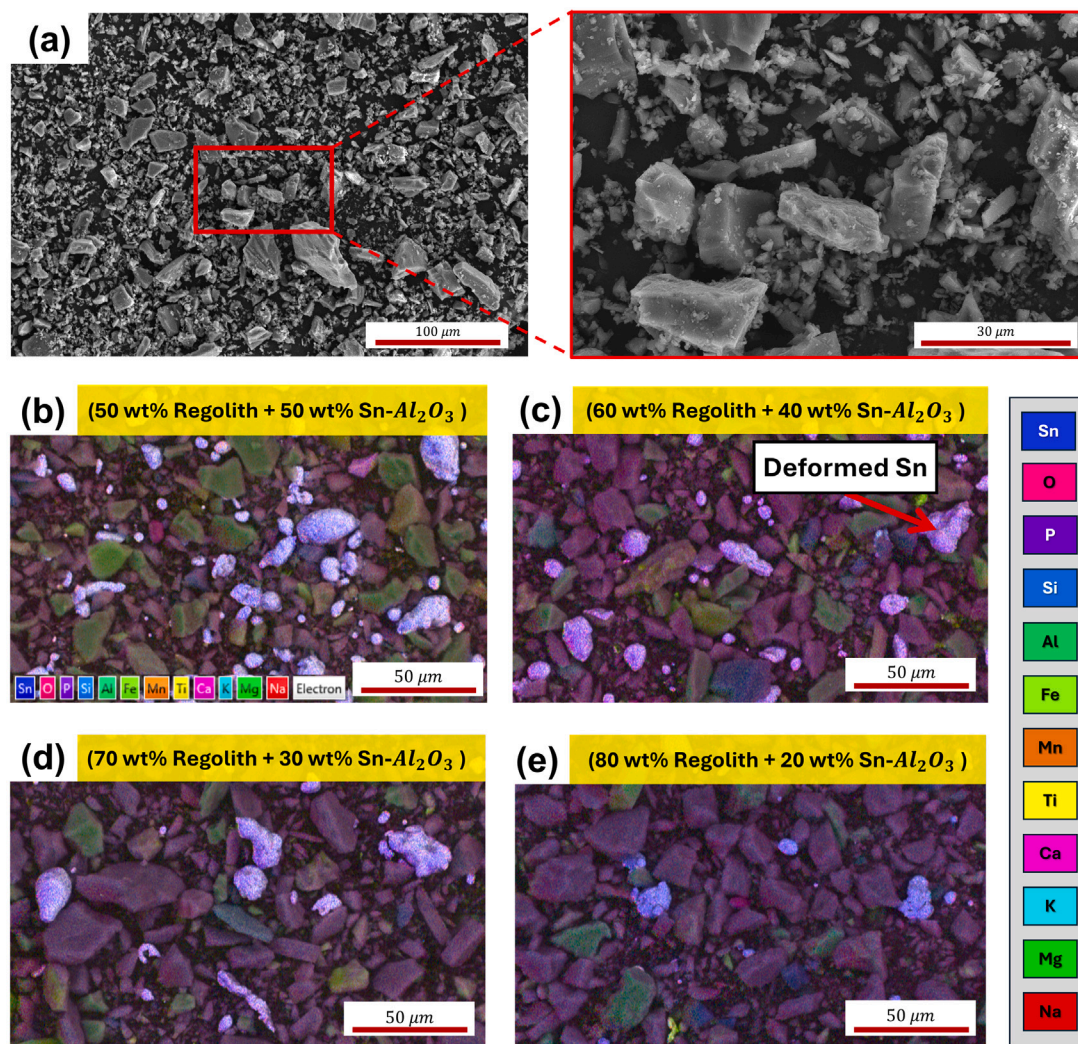


Fig. 2. Morphology and elemental distribution of the synthesized composite powder feedstocks: (a) SEM image of LHS-1D lunar regolith simulant powder; (b–e) EDX elemental maps of regolith–Sn–Al₂O₃ composite powders containing (b) 50 wt%, (c) 60 wt%, (d) 70 wt%, and (e) 80 wt% regolith.

content on the overall coating performance, particularly in terms of electrical conductivity and porosity.

2.3. Substrate

Polytetrafluoroethylene (PTFE, *McMaster*) was employed as the substrate material due to its widespread use in space-related applications [29,30], including thermal insulation, wiring insulation, and sealing components [31,32]. Moreover, PTFE is a highly electronegative material in the triboelectric series, making it an ideal tribo-negative layer for TENGs [33]. Its excellent dielectric properties, chemical stability, and low surface energy further enhance charge generation and retention during contact–separation cycles, thereby improving the overall energy harvesting performance of the TENG system [34,35]. In the CSAM experiments, PTFE coupons (30 mm × 20 mm × 3.2 mm) were used. For the TENG applications discussed in Section 3.3, PTFE substrates with dimensions of 25.4 mm × 25.4 mm × 0.254 mm were employed. In all experiments, the PTFE substrates were used without any prior surface treatment.

2.4. Cold Spray Additive Manufacturing (CSAM)

The composite feedstock powders were deposited onto the PTFE substrate using the CSAM process to fabricate electrically conductive

coatings that serve as TENG electrodes (see Fig. 1b, right panel). CSAM is a solid-state process in which micron-sized powder particles (typically 10–100 μm) are accelerated to supersonic velocities using a pressurized gas stream [20]. Upon impact, the particles undergo severe plastic deformation and adhere to the substrate without melting [36]. This low-temperature solid-state process minimizes oxidation, phase transformations, and residual thermal stresses, making it particularly suitable for depositing temperature-sensitive materials (e.g., PTFE) and fabricating dense, high-integrity coatings [36]. Recently, CSAM has also emerged as a promising approach for the fabrication of reliable and durable TENGs with enhanced performance [8,37–39].

In this study, a robotic low-pressure CSAM system (*Titomic D-523*) was employed for the electrode deposition (see Fig. 1c). The process parameters included a gas pressure (gauge) of 0.5 MPa, and the gas temperature of 300°C, the nozzle traverse speed of 25 mm/s, and a spray standoff distance of 30 mm. These parameters were selected based on our prior work on the deposition of pure lunar regolith onto PTFE substrates [40]. In the present study, the parameter selection was further justified by finite element (FE) simulations of single-particle impact on a PTFE substrate, and to evaluate deformation behavior and bonding conditions. The FE simulation setup and results are provided in Figure S1 (Supporting Information). Under these operating conditions, precise and programmable (direct-write) deposition of the composite regolith feedstock onto the target substrate was achieved. As

demonstrated in Fig. 1d, CSAM enables mask-free and programmable deposition of electrically conductive coatings derived from composite regolith feedstock onto polymeric substrates. Video S1 (Supporting Information) further demonstrates the deposition process, highlighting the capability for robotic-controlled direct-write regolith-based coating and surface functionalization.

2.5. Characterization

The microstructure of the fabricated coatings (electrodes) was studied using scanning electron microscopy (SEM, *FEI Versa*). Energy Dispersive X-ray Spectroscopy (EDX) was used to analyze elemental composition of the coatings. For cross-sectional analysis, specimens were cold-mounted in resin (90% resin, 10% hardener), cured for 24 h, and sequentially polished with abrasive papers (grit 240, 400, 600, 1200), followed by fine polishing. To mitigate charging from the composite powder-coated PTFE samples, a thin platinum (Pt) layer was sputtered prior to SEM and EDX imaging. Coating thickness and porosity were determined from SEM micrographs using *ImageJ* software. The electrical properties of the coatings were evaluated by measuring resistance using a digital multimeter (*Fluke*) via a two-point probe method. Mechanical performance of the specimens was assessed via tensile testing using a universal testing machine (*Instron 4204*) to quantitatively assess coating-induced changes in strength and ductility. Adhesion strength was qualitatively evaluated using a cross-cut test following ASTM D3359 [41]. Furthermore, radiation tests on the coated specimens were conducted using a high-power 60 MeV electron linear accelerator located at the Gaertner LINAC Center at Rensselaer Polytechnic Institute (RPI). The specimens were exposed to a mixed high energy photon/neutron radiation field, followed by a post-irradiation electrical resistance and adhesion assessment. For TENG characterization, a custom-build programmable pneumatic actuator, coupled with an oscilloscope (*Agilent, DSO1014A*), was used to measure the open-circuit voltage (V_{oc}) (see Figure S2 and Video S2, Supporting Information). The short-circuit current (I_{sc}) was obtained with a current amplifier (*Edmund Optics*) interfaced with the oscilloscope. Mechanical loading was applied to the energy devices using a programmable pneumatic actuator operated at various frequencies (i.e., 1 Hz, 2 Hz, 3 Hz).

3. Results and discussion

3.1. Feedstock powder characterization

Fig. 2b–e present the EDX images, illustrating both the particle size and elemental distribution of the synthesized composite powder feedstocks across regolith weight ratios (from 50 to 80 wt%). EDX analyses confirmed the presence of key regolith elements (viz, Sn, Si, Al, Ca, O, Fe, Mg, Ti) within the regolith-metal matrix. Notably, the results indicate that Sn particles undergo significant plastic deformation during ball milling because of repeated high-energy impacts. Owing to its inherent ductility, the Sn phase plastically deforms and locally adheres to the regolith particles, resulting in the formation of a conformal metallic network at the particle interfaces. This behavior promotes particle–particle bonding and improve electrical conduction pathways, both of which are critical for achieving robust deposition and conductive coating formation during the CSAM process.

3.2. Characterization of fabricated coatings (electrodes)

3.2.1. Electrical performance

Fig. 3a shows the variation in electrical resistance of the coatings with various regolith weight ratios (i.e., 50 to 80 wt%). The resistance was measured across the samples with a coating length of 30 mm using the two-point probe method. The results indicate that coatings containing 50, 60, and 70 wt% regolith exhibit measurable electrical conductivity, suggesting the formation of a continuous metallic

network that enables efficient electron transport. SEM images in Fig. 4a–g further confirmed the presence of this interconnected network within the coatings. Moreover, the electrical resistance increased with increasing regolith content. This trend is attributed to the progressive reduction in the conductive metallic phase (Sn), which limits the formation of continuous electron transport pathways. As the regolith fraction increases, the higher concentration of brittle, non-conductive particles introduces greater interfacial resistance and disrupts conductive networks, resulting in increased overall electrical resistance. This behavior is also likely associated with increased porosity at higher regolith contents, as evidenced in Fig. 3b, which further impedes effective electron transport.

Notably, stable conductivity was not observed when the regolith content exceeds 70 wt%, and no measurable conductivity was detected at 80 wt%. As shown in Fig. 4h, the insufficient Sn content at higher regolith loadings prevents the formation of continuous conductive networks, leading to disrupted interparticle bonding and loss of electrical conductivity. This effect is further amplified by the high level of porosity (~65%) (see Fig. 3b), which hinders electron transport by breaking conductive pathways. Furthermore, since conductivity is inherently three-dimensional across the coating thickness, cross-sectional SEM images in Fig. 4b, e, g, and i provided additional insight. Specifically, a well-defined conductive network (i.e., Sn-dominated regions) was observed for the 50 wt% case (Fig. 4c), whereas such interconnected pathways were progressively diminished at 80 wt% regolith content (Fig. 4j).

Fig. 3c presents the electrical conductivity and coating thickness as a function of regolith weight fraction. The conductivity was calculated using Eq. (1), where σ is the electrical conductivity ($S\ m^{-1}$), ρ is the electrical resistivity ($\Omega\ m$), L is the coating length (m), R is the measured electrical resistance (Ω), and $A = w \times t$ is the cross-sectional area (m^2), where w and t denote the coating width (m) and thickness (m), respectively. The coating thickness (t) was determined from cross-sectional SEM images (see Fig. 4b, e, g, and i). As shown in Fig. 3c, the coating thickness decreased with increasing regolith fraction (i.e., from ~39 μm at 50 wt% regolith to ~16 μm at 80 wt% regolith). This trend can be attributed to the reduced content of the ductile metallic phase (Sn). Higher regolith loading limits plastic deformation upon impact, resulting in decreased deposition efficiency and, consequently, thinner coatings. In turn, the reduced coating thickness at higher regolith fractions results in lower electrical conductivity (i.e., higher resistance), as reflected by the relationship in Eq. (1).

$$\sigma = \frac{1}{\rho} = \frac{L}{RA} = \frac{L}{R(w \times t)} \quad (1)$$

Collectively, the results demonstrate the viability of fabricating conductive coatings on polymeric substrates (e.g., PTFE) using engineered regolith-metal composite feedstock powders via solid-state CSAM. The findings further revealed that the electrical performance of the lunar-regolith-metal matrix coatings is strictly governed by the regolith content. This highlights the importance of engineering the regolith-metal ratio to achieve desired, continuous electron transport pathways. In this study, the highest electrical conductivity was achieved at a 50 wt% regolith fraction (i.e., 50–50 wt% regolith–metal mixture), reaching $\sim 4.9 \times 10^5\ S\ m^{-1}$. This value is promising and is only approximately one order of magnitude lower than the bulk electrical conductivity of Sn ($\sim 9.1 \times 10^6\ S\ m^{-1}$) [38]. The surface EDX maps for this composition in Fig. 4k confirmed the formation of interconnected metallic network, with Sn-rich regions uniformly dispersed and effectively bridging the regolith contents. This distribution facilitates efficient charge transport and enhances overall electrical performance. Such a well-conductive network is highly promising for TENG electrodes, where efficient electron flow and robust interfacial contact are critical for improved energy harvesting performance. Hence, the coatings with 50–50 wt% regolith–metal mixture was selected for subsequent characterization studies and for the fabrication of TENG devices in the following sections.

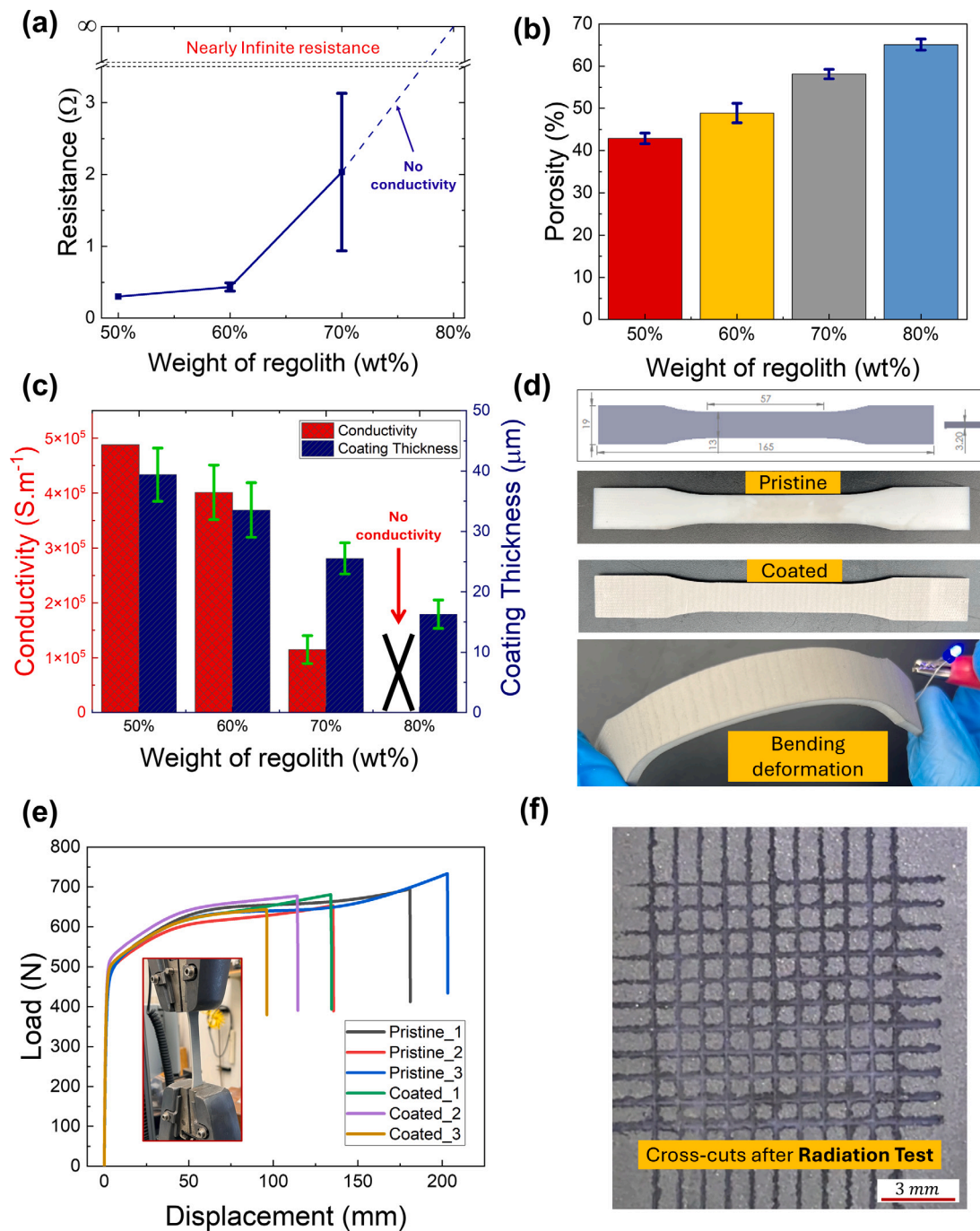


Fig. 3. Characterization of the fabricated coatings (triboelectric layers) : (a) Electrical resistance versus regolith content; (b) Surface porosity versus regolith content; (c) Electrical conductivity and coating thickness as functions of regolith content; (d) Bending durability assessment; (e) Load–displacement responses of pristine and coated samples; (f) Adhesion test specimen after radiation exposure showing surface integrity and coating adhesion stability.

Taken together, these results establish a clear composition–structure–property relationship for CSAM-fabricated regolith composite coatings. Increasing the regolith fraction reduces the relative amount of ductile Sn available for impact-induced deformation and interparticle bonding, resulting in increased porosity, reduced coating thickness, and greater discontinuity of the conductive network. Consequently, electrical conductivity progressively decreases with increasing regolith content and is ultimately lost at sufficiently high regolith fractions. These findings highlight the critical role of the ductile metallic phase in simultaneously promoting deposition, structural cohesion, and electrical connectivity during CSAM.

3.2.2. Mechanical performance

Tensile tests were conducted on dog-bone-shaped specimens in accordance with the ASTM D638 standard [42]. The specimen geometry is illustrated in Fig. 3d (top), with a gauge length of 50 mm. Fig. 3d also presents images of the pristine PTFE sample, the conductive coating-coated PTFE sample, and the coated sample under bending deformation. Notably, the illumination of an LED connected to the bent specimen demonstrates the electrical stability and conductivity retention of the coating under mechanical deformation. Both pristine and coated specimens (50–50 wt% regolith–metal mixture) were tested using a universal testing machine (UTM, Instron 4204) to evaluate their

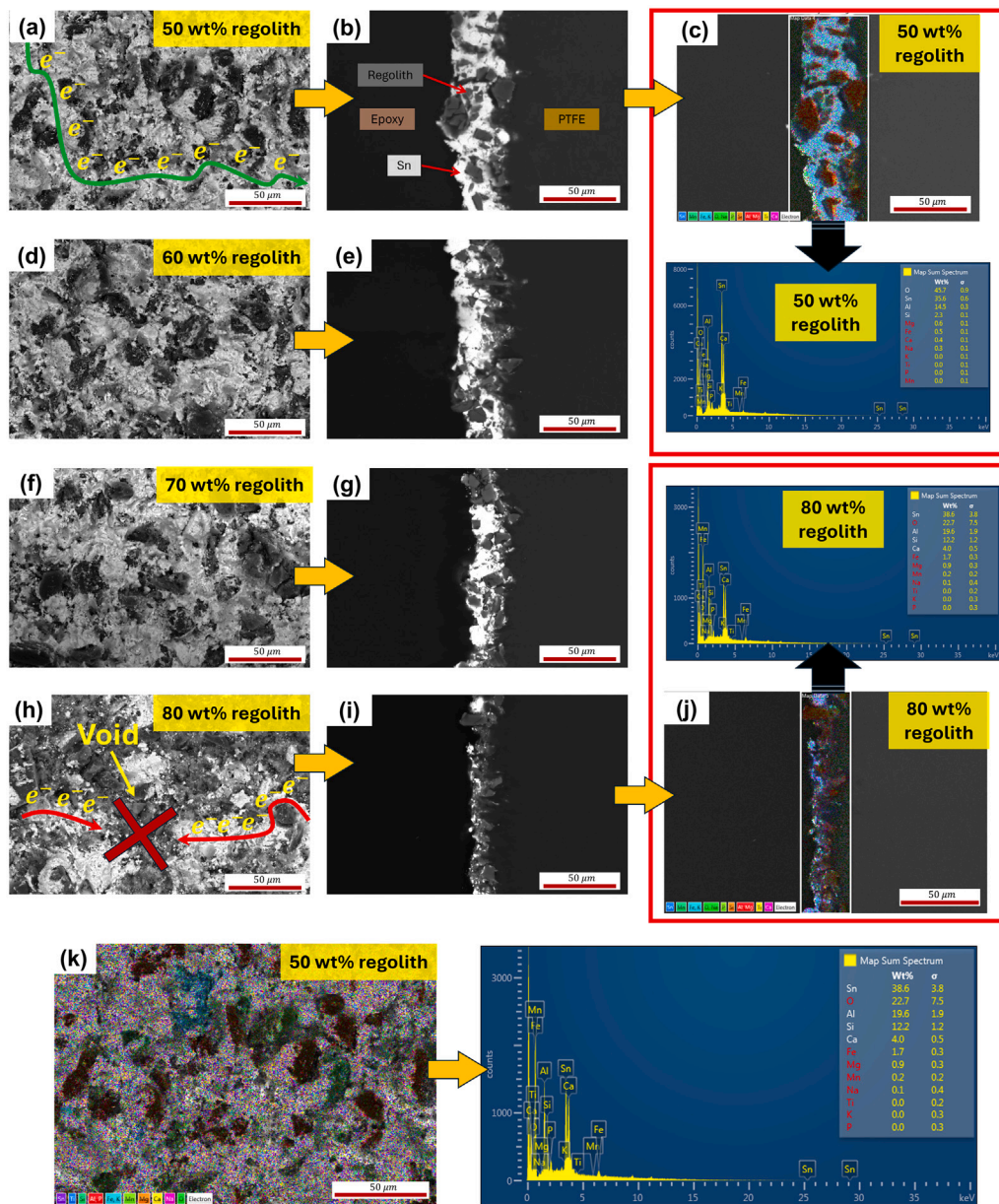


Fig. 4. Microstructural characterization of the fabricated coatings: (a,d,f,h) Surface SEM images of coatings containing 50, 60, 70, and 80 wt% regolith, respectively; (b,e,g,i) Corresponding cross-sectional SEM images; (c,j) Cross-sectional EDX maps and spectra for the 50 and 80 wt% regolith coatings; (k) Surface EDX map and spectrum of the 50 wt% regolith coating.

mechanical performance and assess any potential degradation induced by the CSAM process. The tests were performed under displacement control (350 mm/min), and the applied load was continuously recorded as a function of displacement.

Fig. 3e presents the load–displacement curves of the specimens, while Figure S3 (Supporting Information) summarizes the yield strength, ultimate tensile strength (UTS), and toughness obtained from the experiments. The coated samples exhibit a slight increase in yield strength compared to the pristine (non-coated) PTFE, rising from ~11 MPa for the pristine sample to ~12 MPa for the coated sample (see Figure S3, Supporting Information). This improvement is likely attributed to the presence of the stiff, metallic coating layer, which contributes to load sharing and provides additional constraint at the surface, thereby enhancing the overall resistance to deformation in elastic region.

Conversely, a relative reduction (~3.8%) in UTS was observed for the coated specimens compared to the pristine ones (i.e., UTS = 17 MPa for coated and 16 MPa for pristine specimens). Moreover, the toughness of the coated specimens decreased by ~34% relative to the pristine samples. These reductions are likely attributed to the presence of the relatively brittle coating layer and associated interfacial stress concentrations, which can promote earlier crack initiation and limit plastic deformation. As such, although the CSAM process enables the fabrication of electrically conductive coatings from engineered lunar regolith powder feedstock, the observed reduction in toughness should be carefully considered. This limitation may restrict its applicability in scenarios where high energy absorption or mechanical resilience is required. Despite this relative compromise, the coating remained intact throughout the tests up to specimen failure. Fig. 3d (bottom panel) confirms the robustness of the coating, demonstrating sustained

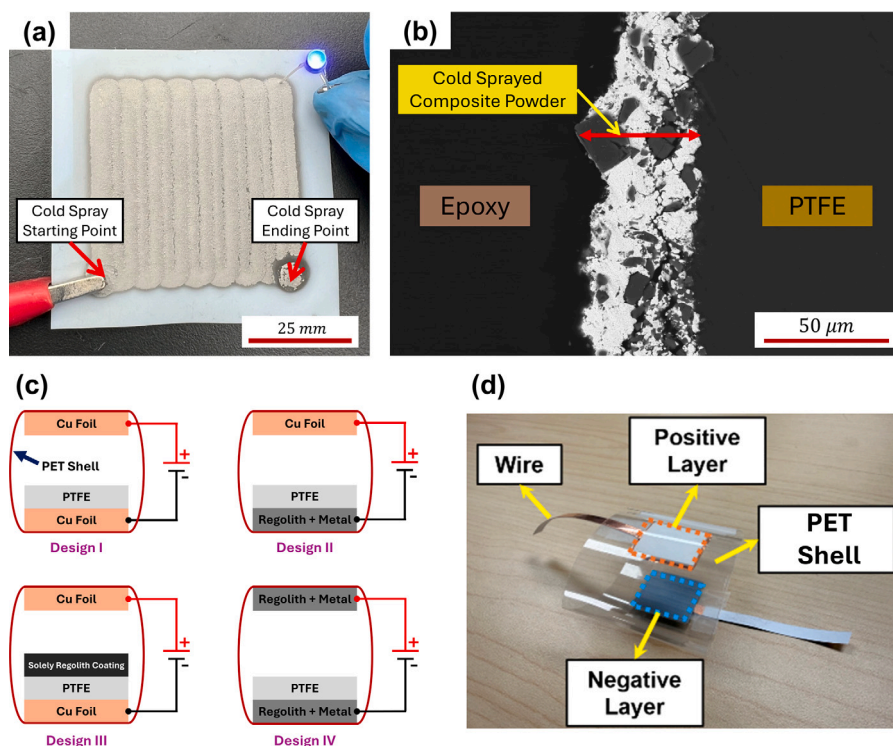


Fig. 5. Fabrication and configuration of the CSAM-based TENG devices: (a) Representative image of a CSAM-fabricated triboelectric layer deposited on a PTFE substrate; (b) Corresponding cross-sectional SEM image; (c) Schematic illustration of the TENG configurations; (d) Representative image of the fabricated lunar regolith-based TENG.

electrical conductivity under severe bending deformation without any signs of delamination or electrical failure.

3.2.3. Radiation resistance

In the radiation tests, an X-ray target was employed with an incident electron beam of 49 MeV and beam powers up to 2.5 kW. The electron beam impinged upon a radiation target [43], producing intense photon and neutron radiation. Beam energy measurements were performed before, during, and after irradiation. The beam current was continuously monitored and logged during testing. For each experiment, two PTFE specimens (2 in. $\times$ 2 in. $\times$ 1/8 in.) were used: one coated and one uncoated (pristine). Each sample was cladded in aluminum during irradiation in the event of coating degradation. Dose deposition in each sample was calculated using the MCNP 6.3.0 radiation transport simulation tool [44] with the irradiation geometry setup [43]. The absorbed radiation dose within the coating was calculated to be 112 kGy under the applied irradiation conditions with an estimated uncertainty of 20%. Recent measurements of charged and neutron particle radiation on the lunar surface report a dose rate of 13.2 ± 1 μ Gy/hour in silicon, though neutron dose rates can increase depending on the depth in the lunar regolith [45]. Notably, the experimental total integrated dose (112 kGy) is significantly higher than the reported radiation dose on the lunar surface integrated over 50 years (0.006 kGy), highlighting the promising radiation tolerance of the fabricated coatings for extreme lunar and space environments. Furthermore, the coatings remained intact after irradiation (Fig. 3f), with no significant change in electrical resistance (~ 0.3 Ω). These findings confirm the excellent radiation resistance and electrical stability of the fabricated coatings under the investigated exposure conditions. It should be noted that preliminary testing was conducted with alternative coating deposition for a similar total integrated dose, where severe coating degradation was observed. While the CSAM-enabled coating performed well and is discussed here, the preliminary testing indicated the need for radiation-based validation of lifetime coating performance on the lunar surface.

3.2.4. Adhesion performance

To evaluate the interfacial adhesion between the conductive coating (electrode) and the underlying PTFE substrate, the cross-cut adhesion test was performed in accordance with ASTM D3359 [41], which provides a qualitative assessment of coating adhesion. In this context, a lattice pattern consisting of a series of parallel and perpendicular cuts, spaced 1 mm apart, was introduced on the coated surface using a sharp blade (see Figure S4a, Supporting Information), ensuring that the cuts penetrated through the coating to the substrate. Subsequently, a pressure-sensitive adhesive tape was firmly applied over the cross-cut area, pressed to ensure uniform contact, and then rapidly peeled off at a consistent angle (Figure S4b). The adhesion performance was assessed based on the extent of coating removal within the lattice grid, following the ASTM D3359 classification scale (e.g., 0B to 5B, see Figure S4c), where 5B indicates no coating detachment and 0B represents severe delamination ASTM D3359 [41]. Fig. 3f shows the post-test inspection under magnification, which revealed no noticeable peeling, flaking, or delamination of the coating from the substrate. According to the ASTM D3359 adhesion scale (Figure S4c, Supporting Information), the coating achieved a rating between 4B–5B, indicating strong adhesion.

3.3. Lunar regolith-based TENGs

3.3.1. Fabrication of TENGs

For the fabrication of TENG electrodes, a regolith–metal composite feedstock (50 wt% regolith – 50 wt% Sn–Al₂O₃) was cold-sprayed onto PTFE films with a thickness of 0.01 in. (~ 254 μ m). Fig. 5a shows the resulting coating deposited on the PTFE film, while the illumination of an LED confirms the electrical conductivity of the coating. The specimen was then cut into 25.4 mm $\times$ 25.4 mm (i.e., 1 in. $\times$ 1 in.) to serve as the triboelectric layer and back electrode for the TENG devices. The average regolith+Sn coating thickness was determined to be 42 ± 4 μ m (see Fig. 5b). For a consistent baseline comparison, Cu foil with a comparable thickness (~ 50 μ m) was used for the reference TENG devices.

As shown in Fig. 5c, four TENG configurations were designed and fabricated to investigate the influence of tribo-pair selection and electrode architecture on device performance. Designs I–IV were constructed by systematically varying the tribo-positive and tribo-negative layers, along with their corresponding back-electrode materials. A representative image of the TENG configuration (Design II) is shown in Fig. 5d, where polyethylene terephthalate (PET) was used as the shell material for each TENG due to its elasticity, enabling cyclic loading of the TENG device. All devices operate in contact–separation mode, in which periodic mechanical excitation induces interfacial charge transfer, thereby generating an alternating electrical potential across the external circuit.

In all configurations, PTFE served as the negative tribolayer. However, in Design III, a pure regolith coating (i.e., without ball milling) was deposited onto the PTFE surface to evaluate the tribo-negative behavior and charge-generation capability of the lunar regolith layer. In Designs I–III, conventional Cu foil was used as the tribo-positive layer and simultaneously served as the back electrode in Designs I and III. In contrast, Designs II and IV employed a regolith–metal composite coating as the back electrode to evaluate its performance relative to Cu foil. Design IV adopted a fully regolith–composite architecture, in which a regolith–metal composite coating served as both the tribo-positive layer and the back electrode. Collectively, these configurations enabled a systematic investigation of the effects of triboelectric pair selection and electrode architecture on interfacial charge transfer and overall TENG performance.

3.3.2. Performance evaluation of TENGs

The electrical output characteristics of the TENGs are presented in Fig. 6a–i. As shown in Fig. 6a, the open-circuit voltage (V_{oc}) measured at 40 psi pressure and 2 Hz reveals clear performance differences driven by the tribo-pair selection and electrode architecture. In detail, Design I, serving as the control, generated a maximum V_{oc} of 37.6 V. Designs III and IV exhibited moderately improved outputs of 48.8 V and 84 V, respectively. Among all configurations, Design II achieved the highest V_{oc} of 142 V, corresponding to a 3.8-fold enhancement over the control TENG (Design I). These results indicate that the regolith–metal composite coating outperformed conventional Cu foil when used as the back-electrode. The superior performance of Design II is attributed to the strong interfacial adhesion and continuous conductive pathways provided by the CSAM-fabricated composite electrode, which facilitate efficient charge collection and transport while minimizing interfacial losses. However, TENG output is also governed by charge generation and retention at the triboelectric interface. The heterogeneous composition and surface characteristics of the regolith-containing layer may influence local dielectric properties, surface states, electron affinity, and charge-trapping behavior, potentially contributing to the observed enhancement. As such, future work should characterize surface charge density, dielectric properties, surface potential, and charge-trapping behavior to elucidate the underlying mechanisms governing the enhanced TENG performance. Collectively, the regolith-based TENG designs II–IV (Fig. 5c) produced significantly higher V_{oc} than the baseline Design I, which contained no regolith (see Fig. 5c). The V_{oc} increased by ~3.8-fold in Design II, ~1.3-fold in Design III, and ~2.2-fold in Design IV compared to the baseline.

In contrast, Design III incorporates a PTFE layer coated with pure regolith powder as the negative tribo-layer, resulting in the lowest performance among all configurations. This behavior is likely attributed to its interfacial contact characteristics, which may reduce effective charge generation and retention, thereby leading to a lower V_{oc} . Design IV, on the other hand, employs a regolith–metal composite as the tribo-positive layer in contact with PTFE, along with a regolith–metal composite back electrode. While Design IV outperformed Design I and III, its performance remained inferior to that of Design II. Although composite layers (regolith + metal) exhibit relatively low surface resistance (Fig. 3a) and forms a continuous conductive network (Fig. 4a),

its interfacial chemistry and charge-transfer characteristics may differ from those of Cu foil used as the tribo-positive layer. In particular, the heterogeneous nature of the composite surface, consisting of insulating regolith particles embedded within a metallic matrix, can reduce the effective contact electrification efficiency by limiting the availability of active charge-transfer sites [46,47]. Furthermore, the presence of dielectric constituents may introduce charge trapping and screening effects, thereby suppressing net charge density generation [48].

Taken together, these findings reveal that although each configuration was capable of generating voltage, the use of a regolith–metal composite as the back electrode provides a more effective conductive pathway and enhances the overall electrical performance of the TENG device. Notably, the regolith–metal composite coating can still serve as both the tribo-positive layer and back electrode when moderate TENG performance with a high degree of regolith utilization is desired. Overall, the combination of Cu foil as the tribo-positive layer, PTFE as the tribo-negative layer, and a mechanically robust regolith+Sn composite as the back electrode in Design II provides the most favorable conditions for charge generation, separation, and transport, resulting in the highest electrical output among all configurations. Accordingly, Design II was selected for further characterization studies.

Fig. 6b shows the V_{oc} of Design II under applied pressures ranging from 20 to 40 psi at a fixed frequency of 2 Hz, exhibiting a monotonic increase from 52.8 V to 88 V and 142 V. This trend is consistent with improved interfacial contact and enhanced charge transfer under higher compressive loading [49]. Frequency-dependent measurements (Fig. 6c,f) further indicate that both V_{oc} and short-circuit current (I_{sc}) increase with load frequency. Specifically, V_{oc} raised from 60 V to 80 V and 142 V, while I_{sc} increased from 8.4 μ A to 15.6 μ A and 17.6 μ A at 0.5, 1.0, and 2.0 Hz, respectively.

To assess durability, the Design II device was subjected to 5000 continuous cycles at 2 Hz and 40 psi (Fig. 6d,e). The I_{sc} output remained stable throughout the test, confirming the mechanical robustness and long-term operational reliability of the regolith+metal-integrated electrode. To evaluate power delivery capability, the device (Design II) was connected to a full-bridge rectifier and tested across external resistive loads ranging from 10 k Ω to 100 M Ω (Fig. 6g,h). As expected from Ohm's law, the average DC voltage increases with load resistance, while the current density decreases [50]. A pronounced rise in power density is observed between 1.0 and 5.0 M Ω , where the external load approaches the internal impedance of the TENG. The power density ($P = V^2/R$) reaches a maximum of 81.3 μ W/cm² at 5.0 M Ω , after which it declines (see Fig. 6h), consistent with the maximum power transfer theorem [51].

Besides, the energy storage capability of Design II was evaluated by charging capacitors of 1.0, 2.2, 4.4, and 10.0 μ F for 60 s through a full-wave bridge rectifier (Fig. 6i). As the capacitance increases, the charging rate decreases, yielding maximum voltages of 2.39 V, 2.09 V, 1.22 V, and 0.49 V, respectively. These results demonstrate the practical feasibility of integrating the Design II TENG into self-powered systems for low-power electronics and intermittent energy-harvesting applications.

To further evaluate the influence of CSAM processing conditions on device performance, regolith coatings were fabricated at varying gas pressures (i.e., 0.4, 0.5, 0.6 MPa), resulting in different coating thicknesses (see Figures S5a–b, Supporting Information). Increasing the gas pressure produced thicker coatings (Figure S5b, Supporting Information) due to enhanced particle impact velocity, promoting particle deposition and interlocking. When incorporated as back electrodes in the Design II configuration, thicker coatings exhibited improved TENG performance, as evidenced by higher V_{oc} (Figure S5c, Supporting Information). This enhancement is likely associated with improved interparticle connectivity and the formation of a more continuous conductive network within the electrode, facilitating effective charge collection and transport. However, pressure-induced changes in coating morphology, density, and interfacial characteristics may also contribute to the observed enhancement.

Table 1
Benchmark comparison of lunar regolith-based TENGs.

Contact materials	Fabrication method	Pressure (kPa)	Freq (Hz)	Contact area (cm ²)	V_{oc} (V)	I_{sc} (μA)	Power density (μW/cm ²)	Ref.
Cu (positive) & PTFE/Regolith-metal (negative)	Cold Spray Additive Manufacturing (CSAM)	~276	2	6.45	142	17.6	81.3	This study
Al/Cu (positive) & Regolith/PDMS (negative)	Doctor Blade	4.17	10	6	10.5	2.2	3.0	[18]
Al/PLA (positive) & PDMS/Regolith-PLA (negative)	Fused Deposition Modeling (FDM)	4.67	10	6	17.4	0.96	~3.0	[19]

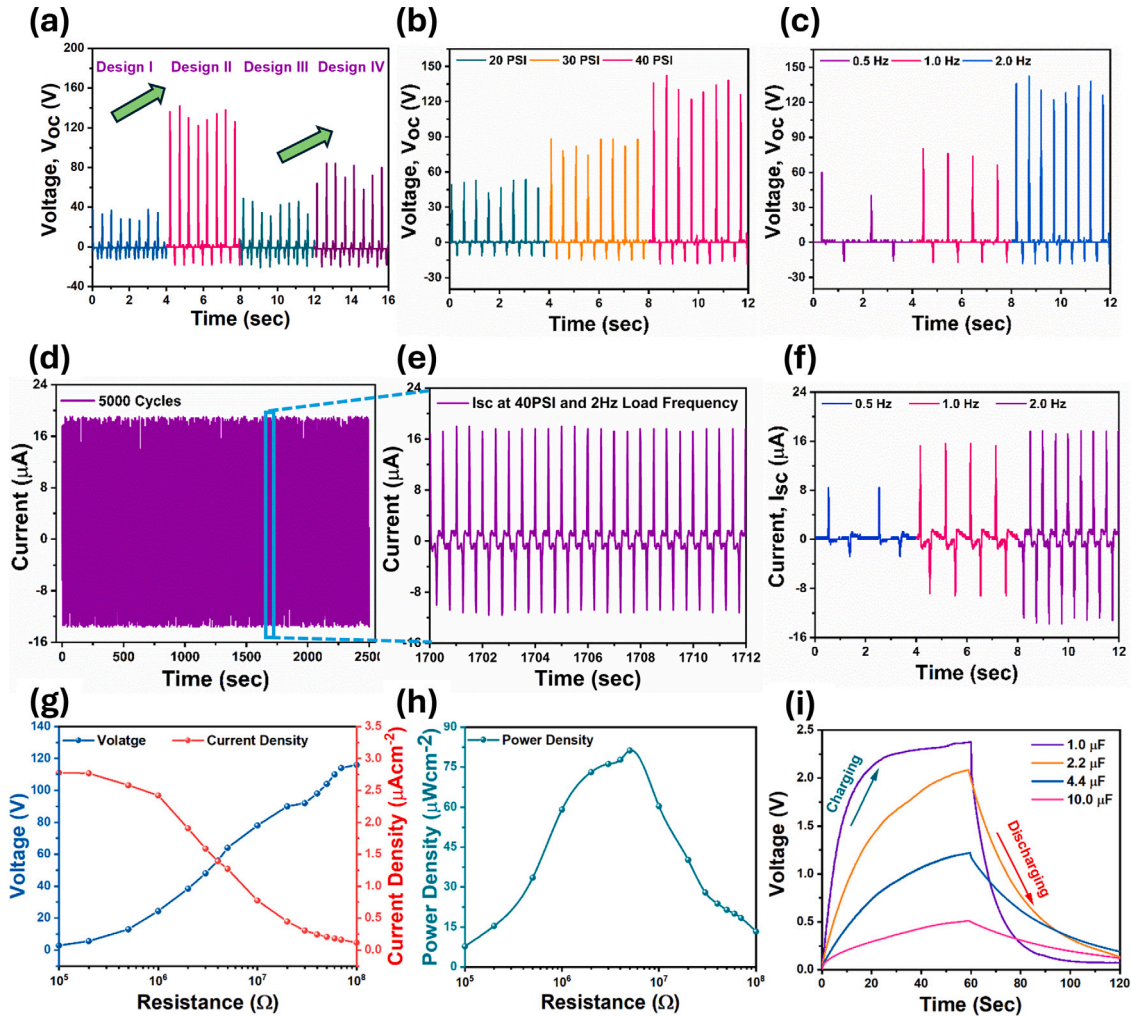


Fig. 6. Characterization of the fabricated TENGs: (a) Open-circuit voltage (V_{oc}) of the four TENG configurations (Design I–IV) at 40 PSI and 2 Hz load frequency; (b) V_{oc} at 20, 30, and 40 PSI at 2 Hz load frequency; (c) V_{oc} at 40 PSI and 0.5–2 Hz load frequency; (d) Durability test results; (e) Magnified image after 5000 cycles; (f) Short-circuit current (I_{sc}) at 40 PSI and 0.5–2 Hz load frequency; (g) Average DC voltage, areal current density; and (h) areal power density for various resistive loads (pressure = 40 PSI; frequency = 2 Hz); (i) Capacitor charging over 60s using 1.0, 2.2, 4.4, and 10 μF capacitors connected to a full-bridge rectifier.

3.3.3. Benchmark comparison of lunar regolith-based TENGs

Table 1 presents a benchmark comparison of regolith-based TENGs reported in the literature to date, highlighting their key performance metrics. Notably, the regolith-based TENG developed in this study demonstrates the most promising overall performance among regolith-based nanoenergy devices reported in the literature. Given that power density is a critical metric for evaluating TENG performance, as it normalizes the influence of device size, the present study achieved

a power density of 81.3 μW/cm². This value is substantially higher (~27-fold) than those reported for existing regolith-based TENGs in the literature (see Table 1). To the best of our knowledge, this represents the highest reported power density for regolith-based TENG devices to date. Furthermore, the TENG developed in this work maintained both structural integrity and stable functional performance under high-pressure cyclic loading conditions (~276 kPa), outperforming previously reported regolith-based TENG designs (see Table 1).

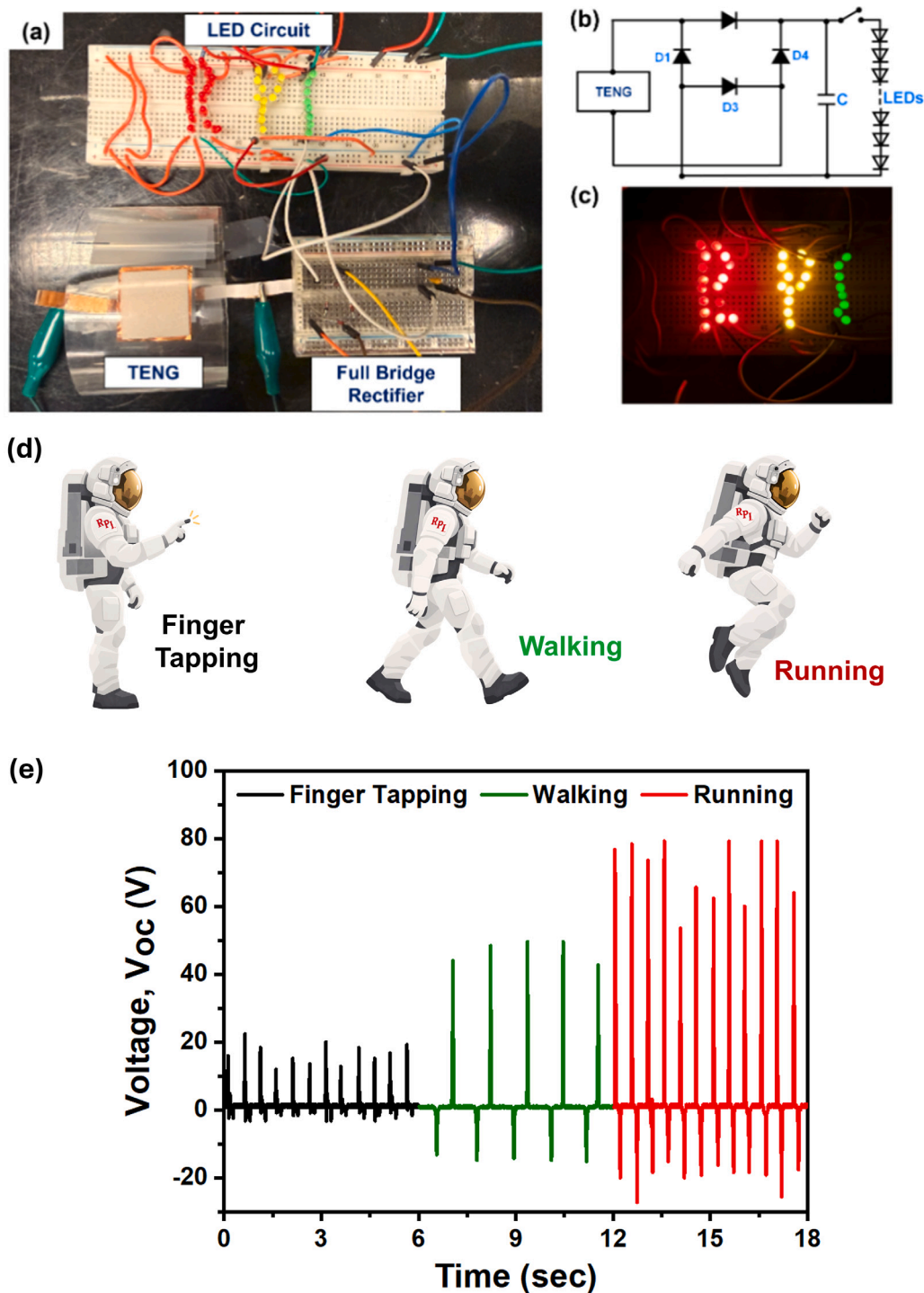


Fig. 7. Energy harvesting and sensing demonstrations using the lunar regolith-based TENG: (a) LED circuit configuration; (b) full-bridge rectifier and capacitor circuit; (c) TENG-powered LED illumination; (d) representative extravehicular astronaut activities; (e) corresponding voltage responses.

3.3.4. Energy harvesting and sensing demonstrations

To demonstrate the practical applicability, the TENG (Design II) was integrated with a full-bridge rectifier and a capacitor, as illustrated in Fig. 7a–b, and employed to power a customized LED circuit comprising 34 LEDs arranged to display the initials of Rensselaer Polytechnic Institute (RPI). As shown in Fig. 7c, the rectified electrical output generated by the TENG was sufficient to illuminate the LED circuit under mechanical excitation, demonstrating effective energy harvesting and power delivery for small electronic devices. Video S3 (Supporting

Information) further demonstrates the real-time operation of the TENG-powered LED circuit. This proof-of-concept demonstration validates the feasibility of using lunar regolith-based TENGs as self-powered energy sources for low-power electronics and sensing applications.

Beyond energy harvesting, the developed TENG was evaluated for motion sensing using representative extravehicular astronaut activities, including finger tapping, walking, and running (Fig. 7d). As shown in Fig. 7e, the TENG exhibited distinct activity-dependent voltage responses, with peak V_{oc} values of approximately 10–20 V for finger tapping, 40–50 V for walking, and up to ~80 V for running. The higher

outputs during walking and running are attributed to greater mechanical excitation, which enhances contact electrification and electrostatic induction. The distinct signal magnitudes demonstrate the potential of the developed TENG as a self-powered motion sensor for differentiating astronaut activities relevant to lunar exploration and extravehicular operations. Collectively, these results highlight the potential of CSAM-fabricated, regolith-based nanoenergy devices for ISRU-enabled energy harvesting and sensing to support future lunar missions.

4. Conclusion

This study investigated the solid-state fabrication of lunar regolith-based TENGs via the CSAM process, demonstrating their potential for sustainable energy harvesting applications. In this context, composite powders consisting of lunar regolith simulant (LHS-1D) and metallic additives (Sn + Al₂O₃) were engineered as feedstock for the CSAM process and subsequently deposited onto PTFE substrate under terrestrial conditions to fabricate TENG electrodes. The principal findings of this work are summarized as follows:

- Electrode (coating) thicknesses of 16–40 µm were achieved with strong adhesion and promising electrical conductivity of up to $4.9 \times 10^5 \text{ S m}^{-1}$.
- The CSAM process did not significantly alter the mechanical properties, while an approximately 9% increase in yield strength was observed compared to the pristine (uncoated) specimen.
- The TENG electrode maintained strong adhesion and electrical conductivity after exposure to a 112 kGy radiation dose, far exceeding the estimated 0.006 kGy accumulated on the lunar surface over 50 years.
- Optimized TENG device (Design II) achieved an V_{oc} of 142 V, I_{sc} of 17.6 µA, and a power density of 81.3 µW/cm², surpassing the performance of previously reported lunar regolith-based TENGs.
- The fabricated TENG device maintained stable performance over 5000 mechanical cycles (i.e., under 40 psi loading at 2 Hz), confirming its reliability and durability.
- The rectified electrical output generated by the TENG successfully powered an LED circuit under mechanical excitation, demonstrating effective energy harvesting and reliable power delivery for small electronic devices.

Collectively, the findings demonstrate the potential of CSAM-fabricated lunar regolith-based TENGs for energy harvesting and self-powered sensing in future lunar missions. While the present study establishes proof-of-concept performance under laboratory conditions, future work should evaluate the CSAM process and TENG performance under lunar-relevant vacuum, thermal, and reduced-gravity conditions. Such environmental validation will be essential for assessing long-term operational reliability and advancing sustainable, ISRU-enabled energy harvesting and sensing systems for future lunar exploration and habitation.

CRedit authorship contribution statement

Sazedur Rahman: Conceptualization, Data curation, Investigation, Methodology, Validation, Visualization, Writing – original draft. **Sk Shamim Hasan Abir:** Investigation, Methodology, Validation, Visualization, Writing – original draft. **Dominik Fritz:** Methodology, Validation, Resources, Writing – original draft. **Johnson Samuel:** Resources, Writing – original draft, Supervision. **Semih Akin:** Conceptualization, Data curation, Investigation, Methodology, Validation, Visualization, Writing – original draft, Resources, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.apmt.2026.103417>.

Data availability

Data will be made available on request.

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